SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Aluri poojitha___________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understandin g of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understandin g with minor gaps	Partial understanding; misses key ideas	Lacks understandin g of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Article Selected

Topic area: Decision Trees / Statistical Learning / Behavioral Anomaly Detection applied to Digital Forensics.

Citation (APA style): Dananjana, W. P., Arambawela, J. S., Gonawala, D. G. S. N., Rathnayaka, R. K. G. H., Senarathne, A. N., & Siriwardena, S. M. D. N. (2025). Machine learning-based criminal behavior analysis for enhanced digital forensics. *PLOS ONE, 20*(10), e0332802. https://doi.org/10.1371/journal.pone.0332802

DOI / Link: https://doi.org/10.1371/journal.pone.0332802

Source: PLOS ONE (peer-reviewed, open access), published 6 October 2025 — satisfies the post-2019 requirement.

Task 1: Article Summary

(a) Citation, Domain and ML Problem

Full citation is given above. The domain is digital forensics and cybercrime investigation, specifically the forensic analysis of web-browser artifacts (browsing history, search queries, cookies, cache/temporary files). The specific ML problem addressed is behavioral anomaly detection: classifying user browsing sessions as normal or suspicious/criminal by learning temporal and cross-platform patterns in browser artifacts, with Decision Tree, Random Forest and Naïve Bayes used as interpretable statistical-learning baselines alongside a primary LSTM-Autoencoder sequence model.

(b) Key Objective and Research Gap (150–200 words)

The authors set out to automate criminal-behavior profiling from digital evidence, an area where traditional forensic analysis is manual, slow, and struggles with today's volume of browser data. Prior digital-forensics literature had applied machine learning mainly to malware and image forensics, while criminal-behavior analysis using browser artifacts remained largely unexplored and reliant on static, single-platform techniques that miss suspects who switch between devices or browsers. Existing behavioral-profiling frameworks (such as Idiographic Digital Profiling and Cyber Behavioral Analysis) offered psychological insight but lacked automated, temporally aware detection, while earlier anomaly-detection studies suffered high false-positive rates and ignored cross-device correlation. To fill this gap, the authors proposed a hybrid model built around a bidirectional LSTM sequence-to-sequence Autoencoder with an attention-enhanced decoder, combined with an ensemble of classical classifiers (Random Forest, Decision Tree, Naïve Bayes, Isolation Forest, One-Class SVM). The model reconstructs normal browsing sequences and flags high reconstruction-error sessions as anomalous, while cross-platform tracking links a suspect's activity across Chrome, Firefox and Edge on both desktop and mobile.

Task 2: Methodology Analysis

(a) ML Techniques and Dataset

Primary technique: a two-layer bidirectional LSTM encoder (128 units/layer) feeding a two-layer LSTM decoder (128 units) with an additive attention mechanism and 0.3 dropout — a Sequence-to-Sequence Autoencoder trained to reconstruct normal browsing sequences, with anomalies flagged when reconstruction error (Mean Absolute Error) exceeds a mean + 3σ threshold tuned per cross-validation fold.

Statistical/tree baselines used for interpretability and ensemble fusion: Random Forest (200 trees, max depth 20, Gini criterion, class-balanced weights), a Decision Tree (max depth 15, min_samples_split = 4, entropy/information-gain splitting criterion) as a simple interpretable baseline, and Gaussian Naïve Bayes (Laplace smoothing, $\alpha = 1.0$). Isolation Forest and One-Class SVM were added as post-reconstruction outlier scorers, and the final anomaly score is a weighted fusion (LSTM-AE 0.5, Random Forest 0.3, Isolation Forest 0.2).

Dataset: the Digital-Forensics v1.0 UMAM-DF dataset from NIST's Computer Forensic Reference Data Sets (CFReDS) portal — 180,000 synthetic, anonymized browser-artifact records (browsing history, cookies, cache, mobile logs) with an 85:15 normal-to-anomalous class distribution. Data was cleaned (duplicate/corrupt-record removal, median/mode imputation), normalized (min–max scaling fit only on training folds), and enriched with engineered features (visit frequency, inter-visit time, cross-artifact consistency checks, cyclic time encodings). The data was split 80/10/10 for train/validation/test, with five-fold stratified cross-validation and nested inner/outer folds used for the final reported metrics.

(b) Mapping to CO4 Topics and Strength/Limitation

- Decision Trees / Inductive Learning: the Decision Tree and Random Forest baselines are direct examples of CO4's decision-tree algorithms, using entropy-based (information-gain) and Gini-based splitting exactly as covered in the course.
- Statistical Learning: Naïve Bayes maps to the course's statistical/probabilistic learning methods (Bayesian classification with Laplace smoothing).
- The LSTM-Autoencoder itself is a deep sequence-learning model rather than classical reinforcement learning, but the overall anomaly-scoring and ensemble-fusion strategy parallels CO4's decision-making/classification theme by combining multiple learners into one classification decision.

Methodological strength: the evaluation is unusually rigorous for an applied forensics paper — five-fold stratified cross-validation, nested folds to avoid data leakage, 95% confidence intervals via DeLong's method and bootstrap resampling, and statistical significance testing between models.

Methodological limitation: the authors themselves acknowledge that cross-device session linking (matching a suspect's activity across phone and desktop) relies on heuristic timestamp/cookie matching with no ground-truth validation, so linkage errors could silently distort the anomaly labels used for training and testing.

Task 3: Results & Critical Evaluation

(a) Key Results

The proposed hybrid model achieved Accuracy 85.4%, Precision 88.1%, Recall 84.7%, F1-Score 86.3%, and AUC 0.93 on held-out test folds, reducing the false-positive rate from 18.9% (an earlier, non-ensemble version of the model) to 10.7%. Table 1 compares it against the traditional baselines the authors benchmarked.

Model / Approach	Accuracy	Precision	Recall	F1-Score	AUC / FPR
Proposed Hybrid LSTM-Autoencoder + Ensemble (RF, Isolation Forest fusion)	85.4% ± 1.8	88.1% ± 2.1	84.7% ± 1.6	86.3% ± 1.9	AUC 0.93; FPR 10.7%
Earlier LSTM-Autoencoder iteration (no attention/ensemble)	79.35%	80.9%	76.8%	78.8%	FPR 18.9%
Random Forest + Isolation Forest	—	—	—	73.5%	FPR > 20%

(baseline)					
GBM + K-Means Clustering (baseline)	—	—	—	70.2%	FPR > 20%

(b) Critical Evaluation of Results

The reported metrics are plausible rather than inflated: 85.4% accuracy with a 10.7% false-positive rate is a realistic outcome for a class-imbalanced (85:15) anomaly-detection task, and the authors report standard deviations and 95% confidence intervals rather than single-run numbers, which supports the results' credibility. However, two biases are worth flagging. First, the UMAM-DF dataset is synthetic and anonymized; synthetic browsing behavior may not fully capture the diversity or subtlety of genuine criminal browsing patterns, so real-world accuracy could differ. Second, the anomaly threshold and Isolation-Forest contamination parameter were tuned using the known anomaly prevalence within each fold — information an investigator would not have in a real, unlabeled case, which the authors themselves note as a limitation. Additional experiments that would strengthen the findings include testing on a real (ethically sourced, anonymized) forensic dataset, an adversarial-evasion test where a simulated suspect deliberately varies browsing patterns to avoid detection, and a same-dataset head-to-head comparison against a plain LSTM-Autoencoder or other modern baselines, which the authors say they were unable to complete in this revision.

(c) Real-World Applicability

The technique is well suited to forensic triage: flagging a small number of high-risk browsing sessions from a very large seized-device dataset so human investigators can focus their limited time on the most suspicious activity, and to insider-threat monitoring in corporate or government IT environments. Scaling it to industry raises several challenges: the LSTM-Autoencoder is computationally heavier than the tree-based baselines, which matters for resource-constrained or real-time deployments; labeled, real-world criminal browsing data is scarce and ethically sensitive to collect; and continuous behavioral monitoring of employees or citizens raises privacy and proportionality concerns that would need clear legal safeguards before deployment in an operational forensic pipeline.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

- Insight 1 — Decision trees remain valuable even alongside deep learning: the paper keeps a simple entropy-based Decision Tree as an interpretable baseline precisely because, unlike the LSTM-Autoencoder, its splits can be directly explained to a court or investigator — reinforcing the course's emphasis on interpretability as a core property of Decision Tree / Inductive Learning methods (CO4).
- Insight 2 — Class imbalance materially changes how statistical learning methods must be configured: the 85:15 normal/anomalous split forced the authors to use class-balanced weights for the Random Forest and Decision Tree and Laplace smoothing for Naïve Bayes, illustrating why the course's statistical-learning topics (Naïve Bayes, weighted classifiers) cannot be applied blindly to real, imbalanced data.
- Insight 3 — Ensemble fusion of multiple learners (rather than one 'best' algorithm) reduced false positives more than any single model, showing that CO4's separate techniques (decision trees, statistical models, sequence models) are complementary in practice rather than competitors, each contributing a different view of the same classification decision.

(b) Proposed Extension

A feasible extension would be to replace the study's reliance on a single synthetic NIST dataset with transfer learning across two data domains: pre-train the LSTM-Autoencoder on the existing UMAM-DF browser-artifact dataset, then fine-tune and evaluate it on a second, independently sourced (and properly anonymized/consented) real-world browser-log dataset, using the Decision Tree and Random Forest baselines as an interpretable sanity check on each domain. This is feasible because the architecture and preprocessing pipeline already exist and only require a new data-ingestion step; the expected impact would be a much stronger, less dataset-specific estimate of real-world accuracy and false-positive rate, directly answering the authors' own acknowledged limitation that their evaluation is confined to one synthetic benchmark.

Reference

Dananjana, W. P., Arambawela, J. S., Gonawala, D. G. S. N., Rathnayaka, R. K. G. H., Senarathne, A. N., & Siriwardena, S. M. D. N. (2025). Machine learning-based criminal behavior analysis for enhanced digital forensics. PLOS ONE, 20(10), e0332802. https://doi.org/10.1371/journal.pone.0332802